

A 3.00-kg object has a velocity $(6\hat{i} - 2\hat{j})$ m/s. What is the net work done on the object if its velocity changes to $(8\hat{i} + 4\hat{j})$ m/s?

☒ a. 70 J

☐ b. 60 J

☐ c. 50 J

☐ d. 40 J

[Clear my choice](#)

Question 2

Not yet answered

Marked out of 1.00

🚩 Flag question

Which of the following equation represents Hamiltonian function?

- ☐ a. $H = L \sum_k p_k q_k - L$
- ☐ b. $\sum_k p_k q_k - D$
- ☐ c. $H = L \sum_k p_k q_k + L$
- ☐ d. $H = L \sum_k p_k q_k$

Question 3

Not yet answered

Marked out of 1.00

 [Flag question](#)

If the sheet of a bakelite is inserted between the plates of an air capacitor, the capacitance will

- ☐ a. Remains unchanged
- ☐ b. Become zero
- ☐ c. Decrease
- ☒ d. Increase

[Clear my choice](#)

[Back](#)

Time left 2:56:45

Question 4

Answer saved

Marked out of 1.00

[Flag question](#)

What law did Ernest Rutherford use to estimate the size of the nucleus?

- ☒ a. Conservation of linear momentum
- ☐ b. Conservation of nucleon number
- ☐ c. Conservation of angular momentum
- ☐ d. Conservation of energy

[Clear my choice](#)[Previous page](#)[Next page](#)**Quiz navigation**

1

2

3

4

5

6

7

8



Time left 2:56:29

Question 5

Not yet answered

Marked out of 1.00

 Flag question

A bar on a hinge starts from rest and rotates with an angular acceleration $\alpha = 10 + 6t$, where α is in rad/s^2 and t is in seconds. What will be the angular velocity in radians per second through which the bar turns in 2.00 s

- ☐ a. 34
- ☒ b. 42
- ☐ c. 32
- ☐ d. 22

[Clear my choice](#)[Previous page](#)[Next page](#)

Quiz navigation

1

2

3

4

5

6

7

8





Physics B.Sc.

[Back](#)

Time left 2:56:10

Question 6

Not yet answered

Marked out of 1.00

[Flag question](#)

In a certain region of space, the electric field is zero. From this fact, what can you conclude about the electric potential in this region?

- ☒ a. It is zero
- ☐ b. It does not vary with position
- ☐ c. It is positive
- ☐ d. It is negative

[Clear my choice](#)[Previous page](#)[Next page](#)



Physics B.Sc.

[Back](#)

Time left 2:55:46

Question 7

Not yet answered

Marked out of 1.00

[Flag question](#)

Which pair of crystal structures have highest packing factor?

- ☐ a. BCC and FCC
- ☐ b. SC and BCC
- ☐ c. FCC and HCP
- ☐ d. SC and HCP

[Previous page](#)[Next page](#)

Quiz navigation



Physics B.Sc.

[Back](#)

Time left 2:55:23

Question 8

Not yet answered

Marked out of 1.00

[Flag question](#)

A dielectric material must be

- ☐ a. Resistor
- ☐ b. Insulator
- ☐ c. Semi-conductor
- ☐ d. Good conductor

[Previous page](#)[Next page](#)

Quiz navigation

[1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#)



Physics B.Sc.

[Back](#)

Time left 2:55:12

Question 9

Not yet answered

Marked out of 1.00

[Flag question](#)

The most common semiconductor among the following substances is:

- ☐ a. Germanium.
- ☐ b. Galena.
- ☐ c. Copper
- ☐ d. Silicon

[Previous page](#)[Next page](#)



Time left 2:55:05

Question 10

Not yet answered

Marked out of 1.00

[Flag question](#)

The probability distribution used for describing the speeds of various particles within a stationary container at a specific temperature is

- ☐ a. Maxwell-Boltzmann Statistics
- ☐ b. Bose-Einstein Statistics
- ☐ c. Fermi-Dirac Statistics
- ☐ d. All of these

[Previous page](#)[Next page](#)**Quiz navigation**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24

[Back](#)

Time left 2:54:42

Question 11

Not yet answered

Marked out of 1.00

[Flag question](#)

What is the probability of throwing 1 or 2 in a dice marked by numbers 1 to 6?

☐ a. $1/6$ ☐ b. 0☐ c. $2/3$ ☒ d. $1/3$ [Clear my choice](#)[Previous page](#)[Next page](#)**Quiz navigation**

1

2

3

4

5

6

7

8





Physics B.Sc.

[Back](#)

Time left 2:54:35

Question 12

Not yet answered

Marked out of 1.00

[Flag question](#)

When a nucleus is divided into its constituents, energy is:

- ☐ a. transformed into visible light
- ☐ b. created from nothing
- ☐ c. absorbed by the nucleus which then breaks it apart
- ☐ d. destroyed into nothing

[Previous page](#)[Next page](#)

Quiz navigation

[1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#)



Physics B.Sc.

[Back](#)

Question 13

Not yet answered

Marked out of 1.00

Flag question

The first law of thermodynamics is the law of

- ☐ a. conservation of mass
- ☐ b. Conservation of heat
- ☐ c. Conservation of momentum
- ☐ d. Conservation of energy

[Previous page](#)[Next page](#)

Quiz navigation



[Back](#)

Time left 2:54:13

Question 14

Not yet answered

Marked out of 1.00

[Flag question](#)

Which of the following is correct in relation to Maxwell's equation in matter?

- ☐ a. If polarization increases, the charge increases
- ☐ b. If polarization increases, the charge decreases
- ☐ c. If polarization increases, the charge remain constant
- ☐ d. All of the above

[Previous page](#)[Next page](#)**Quiz navigation**

1

2

3

4

5

6

7

8

[Back](#)

Time left 2:54:07

Question 14

Not yet answered

Marked out of 1.00

[Flag question](#)

Which of the following is correct in relation to Maxwell's equation in matter?

- ☐ a. If polarization increases, the charge increases
- ☐ b. If polarization increases, the charge decreases
- ☐ c. If polarization increases, the charge remain constant
- ☒ d. All of the above

[Clear my choice](#)[Previous page](#)[Next page](#)**Quiz navigation**



Physics B.Sc.

[Back](#)

Question 15

Not yet answered

Marked out of 1.00

Flag question

A point defect created when an electron leaves its site is termed as:

- ☐ a. Grain boundary
- ☐ b. Vacancy
- ☐ c. Impurity
- ☐ d. dislocation

[Previous page](#)[Next page](#)

Quiz navigation

1

2

3

4

5

6

7

8



[Back](#)

Time left 2:53:51

Question 16

Not yet answered

Marked out of 1.00

[Flag question](#)

Consider a system containing 3 spin $1/2$ particles. Suppose the total energy of the system is $E = \mu B$. The mean magnetic moment of a particle in the system is

- ☐ a. $-1/3\mu$
- ☐ b. $1/3\mu$
- ☐ c. $2/3\mu$
- ☐ d. $-2/3\mu$

[Previous page](#)[Next page](#)**Quiz navigation**



Physics B.Sc.

[Back](#)

Time left 2:53:35

Question 17

Not yet answered

Marked out of 1.00

[Flag question](#)

The momentum of an object is increased by a factor of 4 in magnitude. By what factor is its kinetic energy changed?

- ☐ a. 8
- ☐ b. 2
- ☐ c. 4
- ☒ d. 16

[Clear my choice](#)

[Back](#)

Time left 2:53:27

Question 18

Not yet answered

Marked out of 1.00

[Flag question](#)

An unknown chemical element is presented by the following formula: A_ZX What is the name of index Z?

- ☐ a. Atomic number
- ☐ b. Atomic mass number
- ☐ c. Orbital quantum number
- ☐ d. Principle quantum number

[Previous page](#)[Next page](#)**Quiz navigation**

[Back](#)

Time left 2:53:08

Question 18

Not yet answered

Marked out of 1.00

[Flag question](#)

An unknown chemical element is presented by the following formula: A_ZX What is the name of index Z?

- ☒ a. Atomic number
- ☐ b. Atomic mass number
- ☐ c. Orbital quantum number
- ☐ d. Principle quantum number

[Clear my choice](#)[Previous page](#)[Next page](#)

Quiz navigation





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Flag question

What is the value and unit of permittivity of free space?

- ☐ a. $\epsilon_0 = 6.85 \times 10^{-12} \text{C}^2 / (\text{N} \cdot \text{m}^2)$
- ☐ b. $\epsilon_0 = 9.85 \times 10^{-12} \text{C}^2 / (\text{N} \cdot \text{m}^2)$
- ☐ c. $\epsilon_0 = 8.85 \times 10^{-12} \text{C}^2 / (\text{N} \cdot \text{m}^2)$
- ☐ d. $\epsilon_0 = 6.85 \times 10^{12} \text{C}^2 / (\text{N} \cdot \text{m}^2)$

Previous page

Next page

Quiz navigation

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40

[Back](#)

Time left 2:52:40

Question 20

Not yet answered

Marked out of 1.00

[Flag question](#)

In ionic crystal/solid, electrical conductivity is

- ☐ a. Very high
- ☐ b. Depends on temperature
- ☐ c. Practically zero
- ☐ d. Depends on the material

[Previous page](#)[Next page](#)**Quiz navigation**



Marked out of 1.00

[Flag question](#)

Which of the following is not the correct expression of Hamiltonian quantum oscillator in terms of operator $\hat{a}$ and its Hermitian conjugate

$\hat{a}^\dagger$, when $\hat{a} = \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} + i\hat{p})$ and $\hat{a}^\dagger = \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} - i\hat{p})$

- ☐ a. $\hat{H} = \left(\hat{a}\hat{a}^\dagger + \frac{3}{2}\right)\hbar\omega$
- ☐ b. $\hat{H} = \frac{1}{2}\left(\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}\right)\hbar\omega$
- ☐ c. $\hat{H} = \left(\hat{a}\hat{a}^\dagger + \frac{1}{2}\right)\hbar\omega$
- ☐ d. $\hat{H} = \left(\hat{a}\hat{a}^\dagger - \frac{1}{2}\right)\hbar\omega$

[Previous page](#)[Next page](#)**Quiz navigation**

**Physics B.Sc.**[Back](#)

Time left 2:52:10

Question 22

Not yet answered

Marked out of 1.00

[Flag question](#)

Light of wavelength 580 nm is incident on a slit having a width of 0.300 mm. The viewing screen is 2.00 m from the slit. What is the width of the central bright fringe?

- ☐ a. 10.73 mm
- ☐ b. 7.73 mm
- ☐ c. 5.73 mm
- ☐ d. 14.73 mm

[Previous page](#)[Next page](#)**Quiz navigation**



Physics B.Sc.

[Back](#)

Time left 2:52:01

Question 23

Not yet answered

Marked out of 1.00

[Flag question](#)

An electron and a proton, moving in opposite directions, are accelerated from rest through the same potential difference. Which particle has the longer wavelength?

- ☐ a. The electron
- ☐ b. The proton
- ☐ c. Neither has a wavelength
- ☐ d. Both will have the same wavelength

[Previous page](#)[Next page](#)



Physics B.Sc.

[Back](#)

Time left 2:51:53

Question 24

Not yet answered

Marked out of 1.00

[Flag question](#)

Which phenomenon best supports the theory that matter has a wave nature?

- ☐ a. Photo diffraction
- ☐ b. Electron momentum
- ☐ c. Electron diffraction
- ☐ d. Photon momentum

[Previous page](#)[Next page](#)

Quiz navigation

[Back](#)

Time left 2:51:44

Question 25

Not yet answered

Marked out of 1.00

[Flag question](#)

Which of the following about the gamma ray is true?

- ☐ a. It carries a positive charge
- ☒ b. It has zero rest mass and a neutral charge
- ☐ c. It can be deflected by a magnetic field
- ☐ d. It carries a negative charge

[Clear my choice](#)[Previous page](#)[Next page](#)**Quiz navigation**

1

2

3

4

5

6

7

8



[Back](#)

Time left 2:51:33

Question 26

Not yet answered

Marked out of 1.00

[Flag question](#)

A material with a high dielectric constant:

- ☐ a. Acts to decrease capacitance per unit volume
- ☐ b. Causes a capacitor to become polarized
- ☒ c. Acts to increase capacitance per unit volume
- ☐ d. Has no effect on capacitance

[Clear my choice](#)[Previous page](#)[Next page](#)**Quiz navigation**



Physics B.Sc.

[Back](#)

Time left 2:51:28

Question 27

Not yet answered

Marked out of 1.00

[Flag question](#)

Which of the following statement is correct?

- ☐ a. The current in the discharging capacitor decays exponentially
- ☐ b. The current in the discharging capacitor decreases constantly
- ☐ c. The current in the discharging capacitor grows exponentially
- ☐ d. The current in the discharging capacitor grows linearly

[Previous page](#)[Next page](#)

[Back](#)

Time left 2:51:17

Question 28

Not yet answered

Marked out of 1.00

[Flag question](#)

Which of the following represents
 $\partial L / (\partial q_i) - d/dt (\partial L / (\partial \dot{q}_i)) = 0$?

- ☐ a. Newton's equations of motion
- ☐ b. Lagrange equations of motion
- ☐ c. Hamilton's equations of motion
- ☐ d. None of the above

[Previous page](#)[Next page](#)**Quiz navigation**



Question 29

Not yet answered

Marked out of 1.00

Flag question

A free electron and a free proton are released in identical electric fields. How do the magnitudes of the electric force exerted on the two particles compare?

- ☐ a. It is thousands of times smaller for the electron
- ☐ b. It is millions of times greater for the electron
- ☐ c. It is thousands of times greater for the electron
- ☐ d. They are equal

Previous page

Next page

Quiz navigation

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24

[Back](#)

Time left 2:50:56

Question 30

Not yet answered

Marked out of 1.00

[Flag question](#)

What is the Planck's constant h ?

- ☐ a. An empirical physical constant involved in the calculation of gravitational effects
- ☐ b. A physical constant useful for calculating atomic size
- ☐ c. A physical constant which is the value of magnetic permeability in a classical vacuum
- ☐ d. A physical constant which relates the energy carried by a photon to its frequency

[Previous page](#)[Next page](#)**Quiz navigation**



Time left 2:50:47

Question 31

Not yet answered

Marked out of 1.00

[Flag question](#)

The quantity required to change the temperature of a body of mass and change in temperature is,

- ☐ a. Latent heat of fusion
- ☐ b. Specific heat capacity
- ☒ c. Specific heat
- ☐ d. Latent

[Clear my choice](#)[Previous page](#)[Next page](#)**Quiz navigation**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24



Physics B.Sc.

[Back](#)

Time left 2:50:42

Question 32

Not yet answered

Marked out of 1.00

[Flag question](#)

The threshold frequency of a metal surface is in the violet light region. What type of radiation will cause photoelectrons to be emitted from the metal's surface?

- ☐ a. Red light
- ☐ b. Infrared light
- ☐ c. Radio waves
- ☐ d. Ultraviolet light

[Previous page](#)[Next page](#)**Quiz navigation**

[Back](#)

Time left 2:50:34

Question 33

Not yet answered

Marked out of 1.00

[Flag question](#)

A parallel-plate capacitor is filled with insulating material of dielectric constant ϵ_r . What effect does this have on its capacitance?

- ☐ a. Increasing
- ☐ b. Decreasing
- ☐ c. No effect
- ☐ d. None of the above

[Previous page](#)[Next page](#)**Quiz navigation**[1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#)

[Back](#)

Time left 2:50:22

Question 34

Not yet answered

Marked out of 1.00

[Flag question](#)

In Quantum mechanics, the value of the commutator of $[\hat{x}, \hat{p}]$ is equal to

- ☐ a. $i\hbar$
- ☐ b. $-i\hbar$
- ☐ c. 0
- ☐ d. None

[Previous page](#)[Next page](#)**Quiz navigation**



Physics B.Sc.

[Back](#)

Time left 2:50:14

Question 35

Not yet answered

Marked out of 1.00

[Flag question](#)

Which one of the following statements is correct about entropy?

- ☐ a. Entropy increases for a reversible process.
- ☐ b. Entropy never increases for any process.
- ☐ c. Entropy of a system increases as the number of accessible states of the system increases
- ☐ d. Entropy of a system increases as the number of accessible states of the system increases and Entropy never increases for any process.

[Back](#)

Time left 2:50:01

Question 36

Not yet answered

Marked out of 1.00

[Flag question](#)Hamiltonian H is defined as

- ☐ a. The difference in energy of the system
- ☐ b. The total energy of the system
- ☒ c. The product of energy of the system
- ☐ d. None of the above

[Clear my choice](#)[Previous page](#)[Next page](#)**Quiz navigation**

[Back](#)

Time left 2:49:56

Question 37

Not yet answered

Marked out of 1.00

[Flag question](#)

Which one is not correct about why lasers are used for cutting materials?

- ☐ a. Lasers are cheaper than other cutting materials
- ☐ b. It never gets dull
- ☐ c. Smoother cuts
- ☐ d. Accuracy

[Previous page](#)[Next page](#)**Quiz navigation**

**Question 38**

Not yet answered

Marked out of 1.00

[Flag question](#)

Which event is likely to take place when a photon of energy equal to the difference in energy between two levels is incident in a system?

- ☐ a. Absorption
- ☐ b. Emission
- ☐ c. Absorption and emission
- ☐ d. Neither absorption nor emission

[Previous page](#)[Next page](#)**Quiz navigation**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32

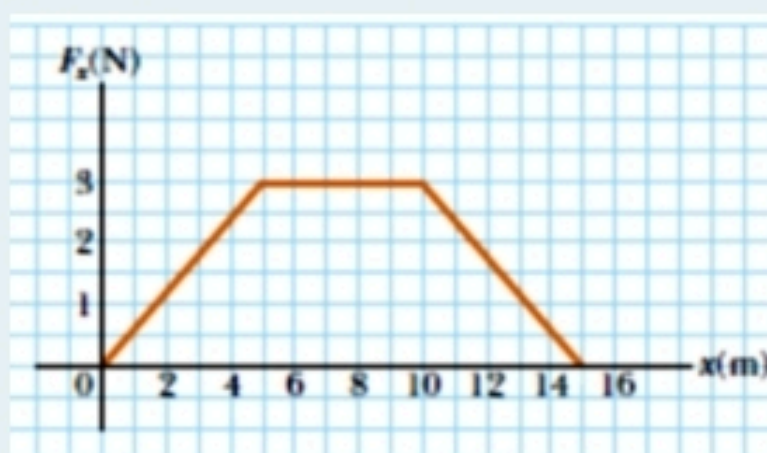
Question 39

Not yet answered

Marked out of 1.00

[Flag question](#)

A particle is subjected to a force that varies with position as shown below. What is the total work done by the force over the distance, $x = 0$ to $x = 15.0$ m



- ☐ a. 30 J
- ☐ b. 40 J
- ☐ c. 35 J
- ☐ d. 25 J

[Previous page](#)[Next page](#)

Quiz navigation

1

2

3

4

5

6

7

8



Time left 2:49:30

Question 40

Not yet answered

Marked out of 1.00

[Flag question](#)

What will be the force acting on a conductor 3m long carrying a current of 50 amperes at right angles to a magnetic field having a flux density of 0.67 tesla?

- ☐ a. 100 N
- ☐ b. 600 N
- ☐ c. 400 N
- ☐ d. 1000 N

[Previous page](#)[Next page](#)**Quiz navigation**

[Back](#)

Time left 2:49:22

Question 41

Not yet answered

Marked out of 1.00

[Flag question](#)

Total virtual work done on N-particle system is

- ☐ a. Zero
- ☐ b. Maximum
- ☐ c. Minimum
- ☐ d. None

[Previous page](#)[Next page](#)**Quiz navigation**



Physics B.Sc.

[Back](#)

Question 42

Not yet answered

Marked out of 1.00

Flag question

A zener diode has

- ☐ a. One pn junction
- ☐ b. Two pn junctions
- ☐ c. Three pn junctions
- ☐ d. Four pn junctions

[Previous page](#)[Next page](#)

Quiz navigation





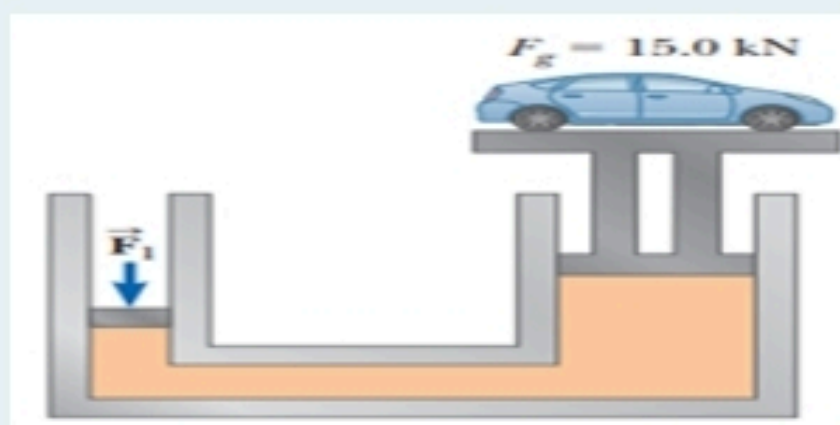
Question 43

Not yet answered

Marked out of 1.00

Flag question

The small piston of a hydraulic lift as shown in the following Figure has a cross-sectional area of 3 cm^2 , and its large piston has a cross-sectional area of 200 cm^2 . What downward force of magnitude F_1 must be applied to the small piston for the lift to raise a load whose weight is $F_g = 1500 \text{ N}$?



- ☐ a. 200 N
- ☐ b. 500 N
- ☐ c. 225 N
- ☐ d. 235 N



Time left 2:48:44

Question 44

Not yet answered

Marked out of 1.00

Flag question

If the ball moves in three dimensions in the space at the position vector

$r = (2t^2\hat{i} + 4t^2\hat{j} + 6k) \text{ m}$, what will be its velocity at $t=3 \text{ s}$?

- ☐ a. $(12\hat{i} + 16\hat{j} + 6k) \text{ m/s}$
- ☐ b. $(18\hat{i} + 36\hat{j} + 6k) \text{ m/s}$
- ☐ c. $(24\hat{i} + 36k) \text{ m/s}$
- ☐ d. $(12\hat{i} + 24\hat{j}) \text{ m/s}$

Previous page

Next page

Quiz navigation



[Back](#)

Time left 2:48:22

Question 45

Not yet answered

Marked out of 1.00

[Flag question](#)

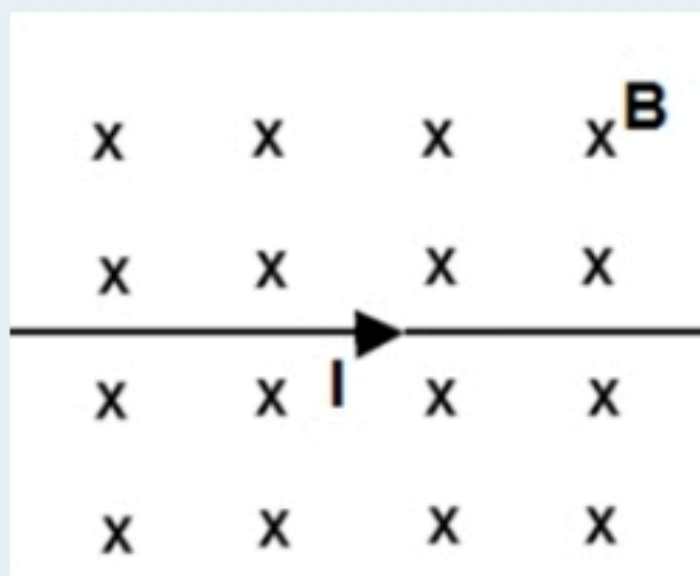
A pan of water is heated from 25°C to 80°C . What is the change in its temperature on the Fahrenheit scale?

- ☐ a. 100°F
- ☐ b. 25°F
- ☐ c. 89°F
- ☐ d. 99°F

[Previous page](#)[Next page](#)**Quiz navigation**



A straight long wire carries an electric current to the right. The current is placed in a uniform magnetic field directed into the page as shown figure below. What is the direction of the magnetic force on the current?



- ☐ a. To the top of the page
- ☐ b. Right
- ☐ c. To the bottom of the page
- ☐ d. Left

[Previous page](#)[Next page](#)**Quiz navigation**



Time left 2:48:00

Question 47

Not yet answered

Marked out of 1.00

[Flag question](#)

The time constant in an R-C circuit may also be defined as the time during which the charging current falls to _____ percent of its initial maximum value.

- ☐ a. 73
- ☐ b. 63
- ☐ c. 42
- ☐ d. 37

[Previous page](#)[Next page](#)**Quiz navigation**



Time left 2:47:51

Question 48

Not yet answered

Marked out of 1.00

Flag question

In black body radiation the peak of wavelength distribution shift to shorter wave lengths as the temperature increases is described by

- ☐ a. Rayleigh-jeans law
- ☐ b. Wien's displacement law
- ☐ c. Stefan's law
- ☐ d. Planck's law

Previous page

Next page

Quiz navigation



Back

Time left 2:47:41

Question 49

Not yet answered

Marked out of 1.00

Flag question

What is the beneficial aspect of nuclear fission?

- ☐ a. The ability to produce more energy than nuclear fusion
- ☐ b. The ability to absorb energy
- ☐ c. There are no beneficial aspects of nuclear fission
- ☐ d. The ability to release tremendous amounts of energy

Previous page

Next page

Quiz navigation





Physics B.Sc.

[Back](#)

Time left 2:47:31

Question 50

Not yet answered

Marked out of 1.00

[Flag question](#)

A coil is wound on iron core which carries current I . The self-induced voltage in the coil is not affected by

- ☐ a. Variation in coil current
- ☒ b. Variation in voltage to the coil
- ☐ c. The resistance of magnetic path
- ☐ d. Change of number of turns of coil

[Clear my choice](#)

[Back](#)**Question 51**

Not yet answered

Marked out of 1.00

Flag question

The line integral of the electric field around any closed path, equals the rate of change of magnetic flux through any surface bounded by that path refers to

- ☐ a. Faraday's law
- ☐ b. Gauss's law in magnetism
- ☐ c. Ampère Maxwell law
- ☐ d. Gauss's law

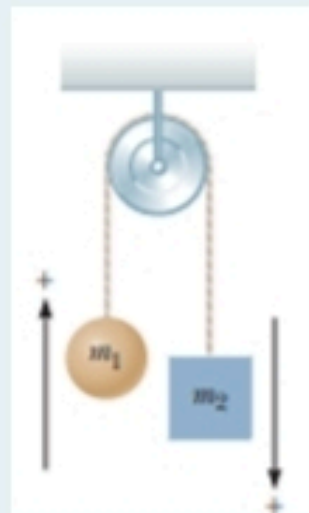
[Previous page](#)[Next page](#)**Quiz navigation**

Plus H • 1199 new messages from 35... in 6h 2

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You KURULUS OSMAN ETHIOPIA pinned "የዜና ቻ..."

Two objects are connected by a string that passes over a frictionless pulley as shown in the following Figure, where m_1, m_2 and a_1 and a_2 are the magnitudes of the respective masses and accelerations. Which mathematical statement is true regarding the magnitude of the acceleration a_2 of the mass m_2 ?



- ☐ a. $a_2 > g$
- ☐ b. $a_2 = g$
- ☐ c. $a_2 < g$
- ☐ d. $a_2 < a_1$



Time left 2:46:56

Question 53

Not yet answered

Marked out of 1.00

[Flag question](#)

Several resistors are connected in series.
Which of the following statements is correct?

- ☐ a. The equivalent resistance is greater than any of the resistances in the group
- ☐ b. The equivalent resistance depends on the voltage applied across the group
- ☐ c. The equivalent resistance is less than any of the resistances in the group
- ☐ d. The equivalent resistance is not equal to the sum of the resistances in the group

[Previous page](#)[Next page](#)**Quiz navigation**

[Back](#)

Time left 2:46:50

Question 54

Not yet answered

Marked out of 1.00

[Flag question](#)

Refraction is used in devices like

- ☐ a. Cameras
- ☐ b. Projectors
- ☐ c. Telescopes
- ☐ d. All the above options

[Previous page](#)[Next page](#)**Quiz navigation**



Physics B.Sc.

[Back](#)

Time left 2:46:40

Question 55

Not yet answered

Marked out of 1.00

[Flag question](#)

When a diode is forward-biased, the anode:

- ☐ a. Alternates between positive and negative relative to the cathode
- ☐ b. Is positive relative to the cathode
- ☐ c. Is negative relative to the cathode
- ☐ d. Is at the same voltage as the cathode

[Previous page](#)[Next page](#)**Quiz navigation**

Physics B.Sc.

[Back](#)

Time left 2:46:26

Question 56

Not yet answered

Marked out of 1.00

[Flag question](#)

Addition of heat at constant pressure to a gas results in

- ☐ a. Raising its volume
- ☐ b. Raising its temperature
- ☐ c. Raising its pressure
- ☐ d. Raising its temperature and doing external work

[Previous page](#)[Next page](#)**Quiz navigation**



Time left 2:46:19

Question 57

Not yet answered

Marked out of 1.00

[Flag question](#)

The number of independent ways in which a mechanical system can move without violating any constraint is called ____.

- ☐ a. Number of freedoms
- ☐ b. Constraint
- ☐ c. Generalized coordinates
- ☐ d. Degrees of freedom

[Previous page](#)[Next page](#)**Quiz navigation**



If a particle is moving along x-axis according to the following equation,

$x(t) = 2t^3 + 4t^2 + 3t$ The magnitude of instantaneous acceleration at $t=2s$ is

- ☐ a. 40 m/s^2
- ☐ b. 24 m/s^2
- ☐ c. 44 m/s^2
- ☐ d. 32 m/s^2

[Previous page](#)[Next page](#)

Quiz navigation

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40



Time left 2:45:55

Question 59

Not yet answered

Marked out of 1.00

Flag question

The correct value for the commutation relations of the components of the orbital angular momentum $\hat{L}_x$ and $\hat{L}_y$ is

- ☐ a. $[\hat{L}_x, \hat{L}_y] = -i\hbar\hat{L}_z$
- ☐ b. $[\hat{L}_x, \hat{L}_y] = i\hbar$
- ☐ c. $[\hat{L}_x, \hat{L}_y] = -i\hbar$
- ☐ d. $[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$

Previous page

Next page

Quiz navigation

1

2

3

4

5

6

7

8



Physics B.Sc.

[Back](#)

Time left 2:45:37

Question 60

Not yet answered

Marked out of 1.00

[Flag question](#)

The dielectric strength of ferroelectric materials is mostly determined by the:

- ☐ a. The presence of magnetic materials nearby
- ☐ b. The frequency of the applied voltage
- ☐ c. The strength of the electric field
- ☐ d. The material's hysteresis loop size

[Previous page](#)[Next page](#)**Quiz navigation**



Time left 2:45:27

Question 61

Not yet answered

Marked out of 1.00

Flag question

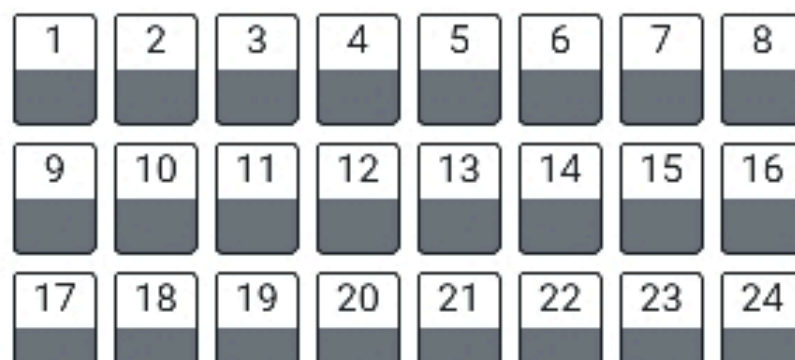
Which of the following is the correct expression of current in an intrinsic semiconductor electronic circuit?

- ☐ a. $I_{\text{total}} = I_{\text{electron}} - I_{\text{hole}}$
- ☐ b. $I_{\text{total}} = I_{\text{electron}} + I_{\text{hole}}$
- ☐ c. $I_{\text{total}} = I_{\text{electron}} + 2I_{\text{hole}}$
- ☐ d. $I_{\text{total}} = 2I_{\text{electron}} + I_{\text{hole}}$

Previous page

Next page

Quiz navigation





Physics B.Sc.

[Back](#)

Time left 2:45:18

Question 62

Not yet answered

Marked out of 1.00

[Flag question](#)

Intensity of a wave is directly proportional to the

- ☐ a. Amplitude
- ☐ b. Cube of amplitude
- ☐ c. Square of amplitude
- ☐ d. Frequency

[Previous page](#)[Next page](#)

Quiz navigation

[1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#)

[Back](#)

Time left 2:45:08

Question 63

Not yet answered

Marked out of 1.00

[Flag question](#)

Which of the assumptions below is not made in the kinetic theory of gases?

- ☐ a. The molecules obey Newton's laws of motion
- ☐ b. The forces between molecules are long range
- ☐ c. The gas is a pure substance
- ☐ d. The number of molecules is very large

[Previous page](#)[Next page](#)**Quiz navigation**



MoEEE

MoE - Exit Exam

Question 04

Not yet answered

Marked out of 1.00

Flag question

Time left 2:45:01

Band gap energy is the energy required by electron to

- ☐ a. Jump from bottom surface of the solid to its top surface
- ☐ b. Move itself from one corner of solid to other corner
- ☐ c. Set free itself from valence band and start conduction
- ☐ d. Jump from center of the solid to its top surface

Previous page

Next page

Quiz navigation

Physics B.Sc.

[Back](#)

Time left 2:44:51

Question 65

Not yet answered

Marked out of 1.00

[Flag question](#)

Which of the following is not an electronic device?

- ☐ a. A computer
- ☐ b. A keyboard
- ☐ c. A mobile
- ☐ d. A magnifying glass

[Previous page](#)[Next page](#)

Quiz navigation

1

2

3

4

5

6

7

8





Physics B.Sc.

[Back](#)

Time left 2:44:43

Question 66

Not yet answered

Marked out of 1.00

[Flag question](#)

The lines of force due to charged particles are

- ☐ a. Always straight
- ☐ b. Sometimes curved
- ☐ c. Always curved
- ☐ d. None of the above

[Previous page](#)[Next page](#)**Quiz navigation**



Physics B.Sc.

[Back](#)

Time left 2:44:36

Question 67

Not yet answered

Marked out of 1.00

[Flag question](#)

The magnetic field of a steady line current describes

- ☐ a. Bio-Savart law
- ☐ b. Gaussian law
- ☐ c. Coulomb's law
- ☐ d. All

[Previous page](#)[Next page](#)

Quiz navigation

1

2

3

4

5

6

7

8



Physics B.Sc.

[Back](#)

Time left 2:44:28

Question 68

Not yet answered

Marked out of 1.00

[Flag question](#)

A collection of molecules that are randomly arranged and held together by weak cohesive forces are

- ☐ a. Solid
- ☐ b. Liquid
- ☐ c. Gas
- ☐ d. Liquid and Gas

[Previous page](#)[Next page](#)**Quiz navigation**



Physics B.Sc.

[Back](#)

Time left 2:44:20

Question 69

Not yet answered

Marked out of 1.00

[Flag question](#)

Classical mechanics describes the motion of _____

- ☐ a. Both Macroscopic and Microscopic object
- ☐ b. Either Macroscopic or Microscopic
- ☐ c. Macroscopic object
- ☐ d. Microscopic object

[Previous page](#)[Next page](#)**Quiz navigation**

**Physics B.Sc.**[Back](#)

Time left 2:44:13

Question 70

Not yet answered

Marked out of 1.00

[Flag question](#)

Modern physics could best be described as the combination of which theories?

- ☐ a. quantum mechanics and classical physics
- ☐ b. Newton's laws of motion and Einstein's theory of relativity
- ☐ c. Newton's laws of motion and classical physics
- ☐ d. quantum mechanics and Einstein's theory of relativity

[Previous page](#)[Next page](#)

Quiz navigation



[Back](#)

Time left 2:44:06

Question 71

Not yet answered

Marked out of 1.00

[Flag question](#)

Which among the following frame of reference has four coordinates: x , y , z , and t ?

- ☐ a. Quantum frame of reference
- ☐ b. 4-dimensional plane
- ☐ c. Inertial frame of reference
- ☐ d. Space-time reference

[Previous page](#)[Next page](#)**Quiz navigation**

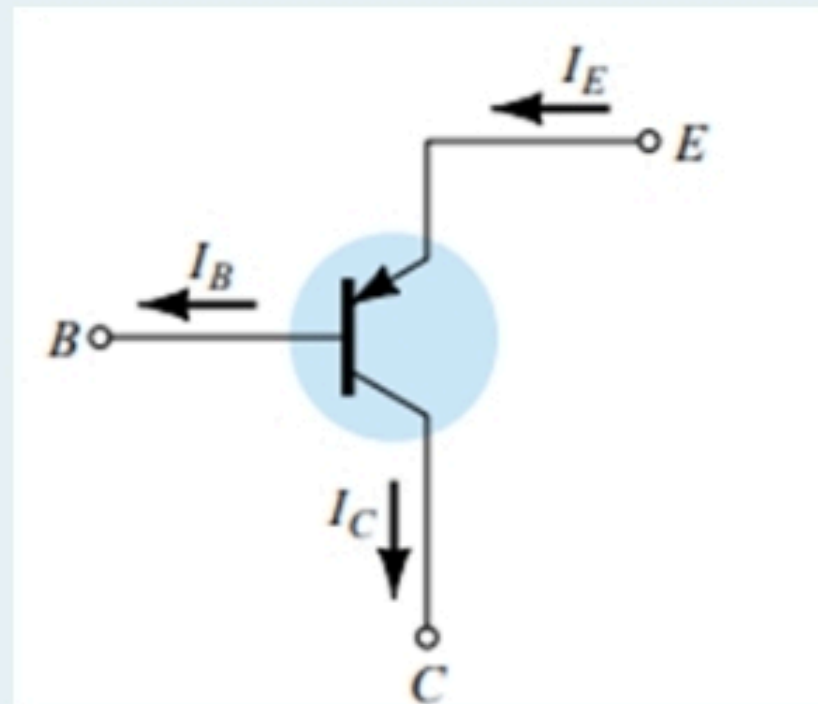
Question 72

Not yet answered

Marked out of 1.00

[Flag question](#)

The notation and symbols used with the common-collector configuration shown in the figure below is



- ☐ a. pnp transistor
- ☐ b. npn transistor
- ☐ c. Doesn't exist
- ☐ d. A and B

[Back](#)

Time left 2:43:33

Question 73

Not yet answered

Marked out of 1.00

[Remove flag](#)

A satellite moves in a circular orbit at a constant speed around the Earth. Which of the following statements is true?

- ☐ a. The satellite moves at constant speed and hence doesn't accelerate
- ☐ b. The satellite has an acceleration directed away from the Earth
- ☐ c. The satellite has an acceleration directed toward the Earth
- ☐ d. No force acts on the satellite

[Previous page](#)[Next page](#)**Quiz navigation**



Physics B.Sc.

[Back](#)

Time left 2:43:23

Question 74

Not yet answered

Marked out of 1.00

[Flag question](#)

Which of the following is not the component of laser?

- ☐ a. Active media
- ☐ b. Resonators
- ☐ c. Pumping mechanism
- ☐ d. The answer is not given

[Previous page](#)[Next page](#)

Quiz navigation



Physics B.Sc.

[Back](#)

Time left 2:43:16

Question 75

Not yet answered

Marked out of 1.00

[Flag question](#)

The _____ in quantum mechanics is analogous to Newton's Laws in classical mechanics.

- ☐ a. Albert Einstein idea of stimulated emission
- ☐ b. Heisenberg uncertainty principle
- ☐ c. Schrödinger equation
- ☐ d. Planck's body radiation

[Previous page](#)[Next page](#)



Physics B.Sc.

[Back](#)

Time left 2:43:08

Question 76

Not yet answered

Marked out of 1.00

[Flag question](#)

Which of the following is not a type of radiation detector?

- ☐ a. Geiger Muller counter
- ☐ b. Proportional counter
- ☐ c. Semiconductor detector
- ☐ d. Flame emission detector

[Previous page](#)[Next page](#)

[Back](#)

Time left 2:38:20

Question 77

Not yet answered

Marked out of 1.00

[Flag question](#)

Which one is false about the classical free electron theory of metals?

- ☐ a. No interaction of electrons with the positive ion cores
- ☐ b. No energy band gap
- ☐ c. Electrons are totally free
- ☐ d. Energy is discrete

[Previous page](#)[Next page](#)**Quiz navigation**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24



Physics B.Sc.

[Back](#)

Time left 2:38:09

Question 78

Not yet answered

Marked out of 1.00

[Flag question](#)

The dominant contribution to current comes from holes in case of which of the following?

- ☐ a. p-type extrinsic semiconductors
- ☐ b. n-type extrinsic semiconductors
- ☐ c. Metals
- ☐ d. Intrinsic semiconductors

[Previous page](#)[Next page](#)



Physics B.Sc.

[Back](#)

Time left 2:37:55

Question 79

Not yet answered

Marked out of 1.00

[Flag question](#)

What is the degree of freedom for plane pendulum which has 3 coordinates and 2 constraints?

- ☐ a. 2
- ☐ b. 5
- ☐ c. 1
- ☐ d. 3



Physics B.Sc.

[Back](#)

Time left 2:37:41

Question 80

Not yet answered

Marked out of 1.00

[Flag question](#)

The coordination number in the case of simple cubic crystal structure is

- ☐ a. 2
- ☐ b. 4
- ☐ c. 6
- ☐ d. 7

[Previous page](#)[Next page](#)

Time left 2:37:21

Question 82

Not yet answered

Marked out of 1.00

[Flag question](#)

A projectile is launched on the Earth with a certain initial velocity and moves without air resistance. Another projectile is launched with the same initial velocity on the Moon, where the acceleration due to gravity is one-sixth as large. How does the maximum altitude of the projectile on the Moon compare with that of the projectile on the Earth?

- ☐ a. It is $\sqrt{6}$ times larger
- ☐ b. It is one-sixth as large
- ☐ c. It is the same
- ☐ d. It is 6 times larger

[Previous page](#)[Next page](#)

Quiz navigation



[Back](#)

Time left 2:37:07

Question 83

Not yet answered

Marked out of 1.00

[Flag question](#)

What is the kinetic energy of an object rotating about a fixed axis with angular momentum, $L = I\omega$?

- ☐ a. $K = I^2\omega$
- ☐ b. $K = L^2 / (2I^2)$
- ☐ c. $K = L^2 / 2I$
- ☐ d. $K = L / 2I$

[Previous page](#)[Next page](#)**Quiz navigation**



Physics B.Sc.

[Back](#)

Time left 2:36:54

Question 84

Not yet answered

Marked out of 1.00

[Flag question](#)

Current changing from 8 A to 12 A in one second induced 20 volts in a coil. The value of inductance is

- ☐ a. 5 H
- ☐ b. 5 mH
- ☐ c. 10 mH
- ☐ d. 10 H

[Previous page](#)[Next page](#)**Quiz navigation**



Physics B.Sc.

[Back](#)

Time left 2:36:47

Question 85

Not yet answered

Marked out of 1.00

[Flag question](#)

All the planets move around the Sun in _____ orbit?

- ☐ a. Parabolic
- ☐ b. Elliptical
- ☐ c. Circular
- ☐ d. Hyperbolic

[Previous page](#)[Next page](#)**Quiz navigation**

1

2

3

4

5

6

7

8



[Back](#)

Time left 2:36:40

Question 86

Not yet answered

Marked out of 1.00

[Flag question](#)

In 3 dimensions, the total number of Bravais lattices and Crystal systems respectively are:

- ☐ a. 14 and 7
- ☐ b. 14 and 6
- ☐ c. 7 and 14
- ☐ d. 8 and 14

[Previous page](#)[Next page](#)**Quiz navigation**

1

2

3

4

5

6

7

8





Physics B.Sc.

[Back](#)

Time left 2:36:32

Question 87

Not yet answered

Marked out of 1.00

[Flag question](#)

When a 4-Kg object is placed on top of a vertical spring, the spring compresses a distance of 2 cm. What is the force constant of the spring?

- ☐ a. 200 N/m
- ☐ b. 2000 N/m
- ☐ c. 400 N/m
- ☐ d. 4000 N/m



Physics B.Sc.

[Back](#)

Time left 2:36:21

Question 88

Not yet answered

Marked out of 1.00

[Flag question](#)

When resistors with different resistances are connected in parallel, which of the following must be the same for each resistor?

- ☐ a. Charge entering each resistor in a given time interval
- ☐ b. Potential difference
- ☐ c. Current
- ☐ d. Power delivered

[Previous page](#)[Next page](#)



Physics B.Sc.

[Back](#)

Time left 2:36:12

Question 89

Not yet answered

Marked out of 1.00

[Flag question](#)

A piece of copper is dropped into a beaker of water. If the water's temperature rises, what happens to the temperature of the copper?

- ☐ a. It increases
- ☐ b. It decreases
- ☐ c. It remains the same
- ☐ d. None of the above

[Previous page](#)[Next page](#)**Quiz navigation**



Physics B.Sc.

[Back](#)

Time left 2:36:02

Question 90

Not yet answered

Marked out of 1.00

[Flag question](#)

What determines the color of light?

- ☐ a. Its intensity and its source
- ☐ b. Its wavelength
- ☐ c. Its source
- ☐ d. Its intensity

[Previous page](#)[Next page](#)

Quiz navigation

[1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#)

[Back](#)

Time left 2:35:54

Question 91

Not yet answered

Marked out of 1.00

[Flag question](#)

Which of the following statement is false?

- ☐ a. The buoyant force is an upward force resulting from an object being partially immersed in a fluid
- ☐ b. The buoyant force is an upward force resulting from an object being wholly immersed in a fluid
- ☐ c. For totally submerged object, the volume of fluid displaced and the object's volume are not equal
- ☐ d. The upward buoyant force is equal to the weight of displaced fluid

[Previous page](#)[Next page](#)**Quiz navigation**

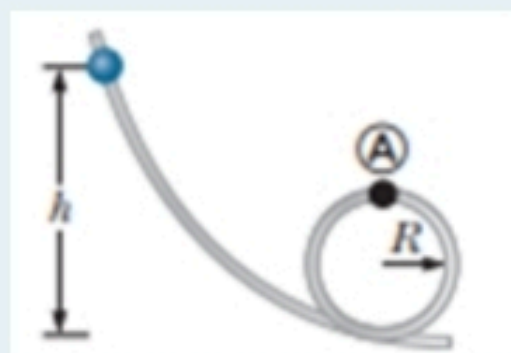
**Question 92**

Not yet answered

Marked out of 1.00

Flag question

A bead slides without friction around a loop-the-loop track as shown in the figure below. The bead is released from rest at a height $h=3.5R$. What is its speed at point A?



- ☐ a. $\sqrt{60R}$
- ☐ b. $\sqrt{40R}$
- ☐ c. $\sqrt{50R}$
- ☐ d. $\sqrt{70R}$

[Previous page](#)[Next page](#)**Quiz navigation**



Physics B.Sc.

[Back](#)

Time left 2:35:33

Question 93

Not yet answered

Marked out of 1.00

[Flag question](#)

Poisson's equation is equal to

- ☐ a. $\nabla \cdot E = \rho/\epsilon_0$
- ☐ b. $\nabla^2 V = -\rho/\epsilon_0$
- ☐ c. $\nabla \cdot E = 0$
- ☐ d. None of the above

[Previous page](#)[Next page](#)

Quiz navigation

[1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#)



Physics B.Sc.

[Back](#)

Time left 2:35:24

Question 94

Not yet answered

Marked out of 1.00

[Flag question](#)

A collection of microstates of a system all having one or more extensive parameters in common is called

- ☐ a. Thermodynamic system
- ☐ b. Statistical ensemble
- ☐ c. Set of microstates
- ☐ d. A macrostate

[Previous page](#)[Next page](#)

Quiz navigation

1

2

3

4

5

6

7

8

Physics B.Sc.

[Back](#)

Time left 2:32:03

Question 95

Not yet answered

Marked out of 1.00

[Flag question](#)

Given the displacement vectors

 $\mathbf{A} = 3\hat{i} - 4\hat{j} + 4\hat{k}$ and $\mathbf{B} = 2\hat{i} + 3\hat{j} - 7\hat{k}$,
the sum of the vectors A and B is

- ☐ a. $\hat{i} - \hat{j} - 3\hat{k}$
- ☐ b. $5\hat{i} + 7\hat{j} + 11\hat{k}$
- ☐ c. $5\hat{i} - \hat{j} - 3\hat{k}$
- ☐ d. $5\hat{i} + 7\hat{j} - 3\hat{k}$

[Previous page](#)[Next page](#)

Quiz navigation

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32



Physics B.Sc.

[Back](#)

Time left 2:30:45

Question 96

Not yet answered

Marked out of 1.00

[Flag question](#)

Schrödinger equation describes about:

- ☐ a. How wave functions change in time
- ☐ b. How electromagnetic waves can physically propagate through space
- ☐ c. How electrons orbit an atom's nucleus
- ☐ d. How electrons are ejected from a metal's surface when radiation falls on it

[Previous page](#)[Next page](#)



Physics B.Sc.

[Back](#)

Time left 2:30:28

Question 97

Not yet answered

Marked out of 1.00

[Flag question](#)

Which phenomenon is best explained by the particle nature of light?

- ☐ a. The Doppler effect
- ☐ b. Interference
- ☐ c. The photoelectric effect
- ☐ d. Polarization

[Previous page](#)[Next page](#)**Quiz navigation**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24



Time left 2:23:42

Question 98

Not yet answered

Marked out of 1.00

Flag question

In quantum mechanics, the operator describing momentum is represented as:

- ☐ a. $\widehat{H}$
- ☐ b. $\hat{r}$
- ☐ c. $\hat{P}$
- ☐ d. $i\hbar \frac{\partial}{\partial t}$

[Previous page](#)[Next page](#)

Quiz navigation

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56



Physics B.Sc.

[Back](#)

Time left 2:23:25

Question 99

Not yet answered

Marked out of 1.00

[Flag question](#)

How much energy is required to raise the temperature of 5.00 kg of lead from 20.0°C to its melting point of 327°C ? The specific heat of lead is $128 \text{ J/kg} \cdot ^{\circ}\text{C}$?

- ☐ a. $1.07 \times 10^5 \text{ J}$
- ☐ b. 8.15×10^4
- ☐ c. $4.04 \times 10^5 \text{ J}$
- ☐ d. $1.96 \times 10^5 \text{ J}$

[Previous page](#)[Next page](#)



Physics B.Sc.

[Back](#)

Time left 2:23:11

Question 100

Not yet answered

Marked out of 1.00

[Flag question](#)

The following are units to measure the effective dose of radiation in a human except:

- ☐ a. RBE
- ☐ b. rad
- ☐ c. rem
- ☐ d. curie

[Previous page](#)[Finish attempt ...](#)**Quiz navigation**